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Metered Verse

Paul Kiparsky

Department of Linguistics, Stanford University, Stanford, California 94305, USA;
email: kiparsky@stanford.edu

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Abstract

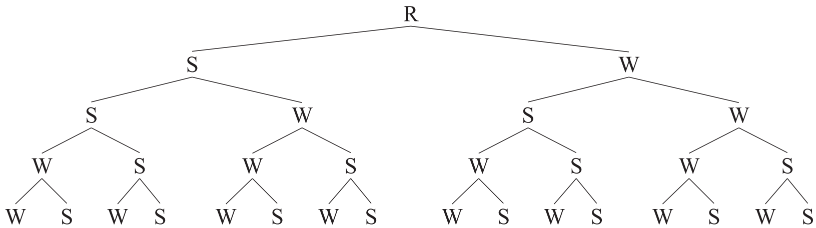
Generative metrics studies versification as a stylization of phonological form. This review article outlines its main goals, hypotheses, and findings and presents a template-matching version of it, which models meters as mappings of abstract verse patterns to their permissible linguistic instantiations. It lays out the key features that verse rhythm shares with rhythm in other cognitive domains and distinguishes it from biological rhythms, offering evidence suggesting that these features are rooted in the language faculty. After a review of the typology of stress-, weight-, and tone-based meters, the predictions of the theory are illustrated in more detail with an analysis of Shakespeare's blank verse. The theory is extended by modeling conventions of setting poems to music as an interface between composition and delivery. English text-setting, which privileges natural phonological stress and phrasing, even at the expense of the poem's meter, lineation, and caesuras, is contrasted with other traditions in which text-setting is more faithful to meter. The negotiation of phonology and meter in song provides a sensitive probe into the prosodic organization of language.

1. RHYTHM AND STRESS

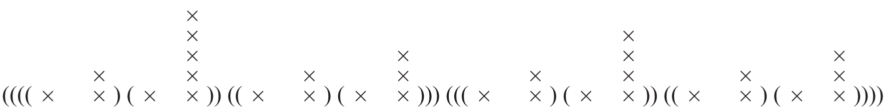
Meter has been called the heartbeat of poetry. But like language itself, and music and dance, it pulsates more intricately than anything in the biological or physical world. While each beat of a healthy heart is identical to the next, meter is based on the regular alternation of prominent and nonprominent beats (stressed/unstressed, strong/weak, downbeat/upbeat).[1] The prominent beats in turn alternate between more prominent and less prominent beats, and so on, up to a unique culmination, generating a prominence hierarchy within a domain (Liberman & Prince 1977). In addition, the prominence peaks at each level of such a hierarchy are grouped with preceding or following less prominent units into measures, usually resolvable into binary constituents in language and meter, but often irreducibly ternary in music and dance. The mind imposes periodicity and constituency even on objectively undifferentiated sequences of beats: We hear the "tick-tick-tick" of a watch as a sequence of "tick-tock" units.

These aspects of rhythmic organization shared by meter and language—alternation, hierarchy, and grouping—can be represented by trees in which each nonterminal node has a unique strong immediate constituent or, equivalently, by bracketed grids where relative prominence is represented by column height (Prince 1983, Hayes 1989):

(1a)



(1b)



The tree representation (example 1a) and the bracketed grid representation (example 1b) convey exactly the same information and are interconvertible. However, they differ somewhat in the formal operations and constraints that can be perspicuously expressed in them.

Certain constituents in the hierarchy have substantive properties of their own. In phonology, these constituents make up the prosodic hierarchy; its stylized counterpart in verse is the metrical hierarchy, which appears to be strictly binary and hence does not map onto the prosodic one:

[1]However, certain pathologies are manifested by binary rhythms between or within heartbeats, as in pulsus alternans, an alternation between strong and weak heartbeats, and dicrotic pulse, with two beats per cardiac cycle, the first on the systole, the second on the diastole. Even the single pulse of a normal heartbeat is actually generated by two contractions—of the atria and ventricles, in that order—which in turn form part of a physiologically complex cardiac cycle. Nature abounds in multiphasic events, but there seems to be no reason to consider them to be alternations between prominent and nonprominent beats. In any case, nothing in nature comes close to the depth of hierarchical rhythmic organization found in the cognitively structured domains of dance, music, verse, and language itself.

(2a) Prosodic hierarchy (in language)	(2b) Metrical hierarchy (in verse)
	Poem
Utterance	Quatrain
Intonation group	Couplet (distich)
Phonological phrase	Line
Word	Half-line (hemistich)
Foot	Dipod
Syllable	Foot
Mora	Beat

The prosodic and metrical hierarchies are subject to STRICT LAYERING, a violable constraint which militates against skipping levels and against improper bracketing [Itô & Mester 2003 (1992)]:

(3) STRICT LAYERING

(3a) A nonterminal unit of the prosodic hierarchy, X^p, is composed of one or more complete units of the immediately lower category, X^{p-1}.

(3b) A nonroot unit of a prosodic hierarchy, X^p, is completely contained in a unit of the immediately superordinate category, X^{p+1}.

A consequence of the grounding of meter in language is that lineation and length restrictions are not defined by counting feet or any other units in example 2b—for phonology cannot count past two. They emerge from constraints on the hierarchical rhythmic structure that versification imposes on texts (Chen 1980, Kiparsky 2006). Accordingly, an iambic pentameter is not simply a sequence of five iambs: It is built from cola that are themselves built from feet. Some of the evidence that confirms this prediction is summarized in Section 3. It follows that the line has no privileged status over the couplet and the stanza, which are often defined by additional metrical constraints on lines, and that lineation is in part conventional, in that any place where an obligatory major prosodic break in the verse regularly divides equivalent units may be considered a line break, by convenience or tradition. 4-3-4-3 ballad quatrains can be printed as fourteener distichs and vice versa, and 3-3-4-3 quatrains are interchangeable with poulter's measure. Some editions of the *Kalevala* print its 8-syllable parallel couplets as single 16-syllable lines, and nobody minds. These considerations argue against the contrary view (Fabb & Halle 2008, p. 242) that lineation and length restrictions are the primary property of verse, and that meter arises as a by-product of counting syllables to fix the length of lines, with rhythm an epiphenomenal "property of the way a sequence of words is read or performed."

In addition to its constitutive structural properties of hierarchy and constituency, meter is enlivened by excursions such as syncopation, skipped beats, doubled beats, and extrametrical beats, followed by a return to the regularly alternating baseline rhythm. A heart that pulsates like verse would have its owner sent, with sirens screaming, to the nearest cardiological ward.

But the complexity is regimented. A meter is defined by a set of constraints on the distribution of prominence-defining phonological features and on the misalignment of linguistic constituents (words, phrases, sentences) with metrical constituents (feet, dipods, cola, lines, stanzas). These constraints can depend on genre and period, and on the level of the hierarchy at which the mismatch occurs. The iambic pentameter of Shakespeare's dramatic verse offers an extraordinarily abundant metrical palette (described in Section 3, below), that of his sonnets less so, but both are metrically richer than Pope's neoclassical heroic couplets, and even that tight form still offers some options denied to most German or Russian verse.

In virtue of being subject to well-defined constraints, metrical verse is the simplest, unmarked form of literary language—indeed, in many unwritten literatures, the first or only form. Prose is a more complex form of literature [as already pointed out by Herder 1960 (1768); see Hanson & Kiparsky 1997], distinguished by rhythmic variety and often by the avoidance of metrical cadences (antimetricality; Borgeson et al. forthcoming).

Metrical verse, lyric as well as epic, is common in preliterate cultures.[2] Even in written literatures, all metrical verse is in principle oral, for writing systems reflect prominence-defining phonological features imperfectly, if at all, and the graphic properties of a text, including spacing and line breaks, are not in themselves constitutive elements of verse but rather are cues to phonological phrasing and lineation. A metrical system not only is a cultural artifact but also is molded within the limits of cognitive and linguistic constraints by functional pressures, of which the most important are that the metrical repertoire should be as expressive as necessary to accommodate the language (fit) and that it should be as restrictive as possible within those bounds (interest) (Hanson & Kiparsky 1996).

The goal of generative metrics is a "grammar" of meter, rather than a "dictionary." It aims to characterize metrical systems as points or regions in a typological space, with linguistic theory the main theoretical model and explanatory engine. Its fundamental hypothesis is that the metrical form of verse is a stylization of phonological form. Appropriately formalized, this hypothesis predicts a typology of possible verse meters which is more narrowly constrained than musical meters in several respects. In particular, verse meter inherits from phonology a strict binarity. Ternary measures, ubiquitous in music, are in verse constructed from binary ones by beat splitting. While polymeters and nonisochronous (additive) meters such as 2-2-3-2-3 or 2-2-2-2-1$\frac{1}{2}$-1$\frac{1}{2}$ are used in musics of Africa and South Asia, they are not found not in their poetries (Deo 2007).

Another consequence of the grounding of meter in language is that lineation and length restrictions are defined not by counting feet or any other units in example 2b—for phonology cannot count past two—but rather by their abstract hierarchical organization. As noted above, an iambic pentameter is not merely a sequence of five iambs: It is built from cola that are themselves built from feet. Some of the evidence that confirms this prediction is summarized in Section 3.

Several approaches to generative metrics are currently being explored (Blumenfeld 2015). One line of investigation which has proved productive treats meter as a mapping of abstract verse patterns ("templates") to their permissible linguistic instantiations. Both templates and their instantiations can be represented by bracketed grids or labeled trees, as in example 1. The abstract verse patterns can be characterized by systems of ranked Optimality-Theoretic markedness constraints, of the same type as the constraints that govern the prosody of language itself. Evidence is mounting that they are very simple and uniform.[3] All the complexity is located in the constraints that delimit the permissible mismatches between abstract metrical patterns and the linguistic patterns of prominence that realize them. These can be formalized by Optimality-Theoretic correspondence constraints. The output of this constraint evaluation is a metrical analysis, which specifies

[2] See Beissinger (2012) for a concise review of the major traditions of oral and oral-derived poetry, with bibliography.

[3] For instance, there is no such thing as "trochaic substitution" in iambs: An inverted initial foot is still an iamb, its inversion licensed by the correspondence rules (Kiparsky 2007).

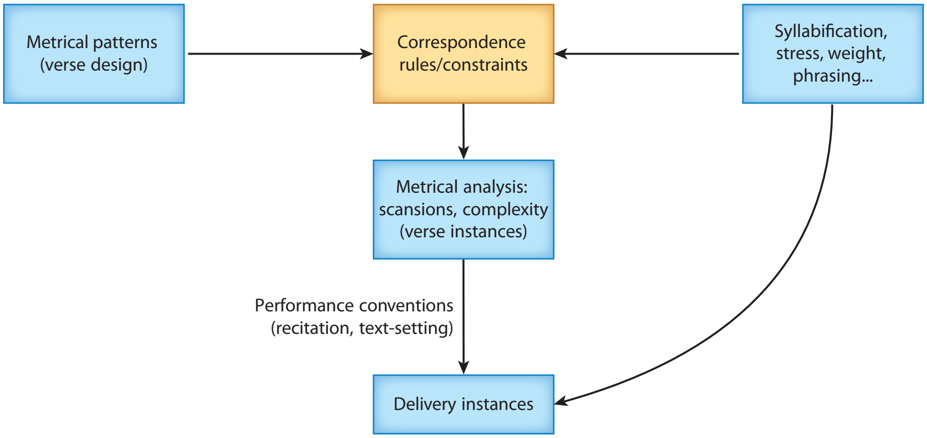


Figure 1

Structure of a metrical theory.

whether a given text instantiates (i.e., legitimately corresponds to) a given abstract metrical pattern and, if so, how complex the instantiation is, as measured by the licensed violations it incurs.

Vocal performance in song, recitation, and drama can profitably be included in the theory, since it is shaped both by meter and by linguistic form, and since prevailing styles of performance historically influence meter. We can model conventions of text-setting as an interface between composition and delivery. **Figure 1** shows the structure of a metrical theory. The constraints not only define the space of metrical variation but also predict the relative frequencies of the permissible variant types (Kiparsky 2006).

According to **Figure 1**, composing and performing a song require matching three hierarchies of alternating prominence: an intrinsic prosodic form assigned by the language's phonology, independently of how it is versified; a metrical parse, independent of how the text is set to music; and a musical rhythm, independent of the words that may be sung to it. The metrical structure of a text is invariant and does not change with the way it is set to music or performed. Each rhythmic tier is representable by a hierarchical tree or grid structure but is subject to its own constraints. The required correspondences and permissible mismatches between them are regulated by conventions that evolve historically within limits grounded in the faculty of language. Predominant metrical systems and recitation, singing, and text-setting practices in a poetic tradition are mutually accommodated, and in time mutually optimized. Performance can be "tilted" to reflect meter, and metrical forms must be compatible with prevailing text-setting and recitation practices. For example, the eighteenth-century practice of reciting verse by highlighting the meter made it impossible to read the work of poets like Wyatt and Donne in a natural way, and demanded poetry that could be metrically parsed with a minimum of mismatches.

The model in **Figure 1** allows for cases in which a literary tradition accesses a different grammar than the ordinary language. For example, metrical practice can be based on the phonology of an earlier stage of the language. In traditional French versification, consonants that are deleted in word-final position count for purposes of rhyme, except that homorganic final voiced and voiceless obstruents are treated as equivalent. For example, *long* and *tronc* rhyme, but neither of them rhymes with *rond* or *pont*, which, however, rhyme with each other; none of them rhymes with *son*. Phonemically and phonetically, all five words end alike: /tʁõ/ [tʁõ], /lõ/ [lõ], /mõ/ [mõ], /ʁõ/ [ʁõ], /sõ/ [sõ]. Morphophonologically, they all end differently: {tronk}, {long}, {pont}, {rond}, {son}—the consonant shows up before suffixes, as in *tronquer*, *longue*, *ponter*, *ronde*, *sonner*. The two rhyming pairs match correctly at the output of the lexical phonology, which Kiparsky (2018a) argues to be a significant level or representation. By final devoicing, both *long* and *tronc* end in /-k/ and both *pont* and *rond* end in /-t/. The evidence that final devoicing takes effect at the word level whereas final deletion is postlexical is that, in the classical *liaison* system (now as old-fashioned as the rhyming convention that reflects it), final voiced stops appear in devoiced form before a following vocalic word in close contact, as in *long hiver* [lɔ̃.ki.vɛʁ] 'long winter' or *grand homme* [grɑ̃.tɔm] 'great man'. Therefore, they must enter the postlexical phonology with the final consonant present in devoiced form. In sum, traditional French versification conventions crucially refer to the lexical representation that is computed by the word phonology and forms the input to the sentence phonology:

(4) Underlying (morphophonemic representation): {tʁonk}, {long}, {pont}, {ʁond}, {son}
Lexical representation: /tʁõk/, /lõk/, /põt/, /ʁõt/, /son/
Phonetic (and structuralist phonemic) representation: [tʁõ], [lõ], [põ], [ʁõ], [sõ]

That such fine points of rhyming outlived the sixteenth-century pronunciation they reflected for several centuries is presumably due to the fact that the living morphophonology of the language kept them intelligible.

2. PROMINENCE

A meter may require prominence in Strong positions, or prohibit prominence in Weak positions, or both. Prominence is expressed by stress, syllable weight, and/or pitch. None of these features is intrinsically binary: Languages can have degrees of stress and weight, and distinctive levels and contours of pitch. The continua are exploited to give texture to verse (Ryan 2017) and, indeed, to literary prose. But remarkably, in categorical constraints they are all binarized—another manifestation of the pervasive binarity of verse structure. Stress meters (syllabotonic meters) work with a binary distinction between stress peaks and nonpeaks. It may be defined on different units in the hierarchy (example 2), each potentially constrained by meter, but always binarily. Quantitative meters, found in classical Greek, Arabic, Sanskrit, Persian, Urdu, Hausa, Somali, and Hungarian, among other languages, establish a binary distinction between light syllables, with one unit of length, or mora, and heavy syllables, with two or more moras. Tone-based meters, such as Chinese regulated verse and Vietnamese Luc Bat, are based on a binary division between the language's lexical tone categories.

Some metrical systems involve interactions of two or even three prominence features: Latin and Skaldic verse obey separate constraints on quantity and stress, and Finnish and Tamil impose special weight conditions on stressed syllables (Ryan 2017). In the *deseterac* meter of Serbian epic songs, stress, weight, and pitch accent all seem to be relevant (Zec 2009), again as binary features.

It appears that the prosodic features that play a role in a language's verse are limited to those which are active in its phonology, either phonemic (distinctive/contrastive) or conditioning processes of word phonology (Kiparsky 2018a). For example, in the phonological systems of Hindi-Urdu, classical Arabic, Finnish, Czech, Latin, and Hungarian, vowel length is distinctive and primary word stress is predictable. Yet only the first two languages rely exclusively on syllable weight in their versification; the others exploit both of these prosodic features to varying extents. This is because word stress plays a crucial role in their respective lexical phonologies, and a negligible one in those of Hindi-Urdu and Arabic. Turkish offers intralinguistic confirmation: The Persian-derived classical meters of Ottoman Turkish are based on syllable weight, while the popular modern meters are based on stress. Correspondingly, vowel length is distinctive only in the Persian-derived (ultimately Arabic) vocabulary that the classical Turkish meters use. English, German, and Swedish meters are almost exclusively stress based, though syllable weight interacts with their lexical phonologies in ways that are reflected in some of their meters. Russian has only stress, and consequently only stress-based meters, with syllable weight playing no discernible role. Mordvin has neither lexical stress nor vowel quantity nor distinctive pitch, and its meters are consequently syllable counting with variable line divisions as the principal rhythmic device (and tend to compensate by rich nonmetrical devices such as rhyme, phrasing, and parallelism). French stress is basically a phrase-level feature, and its meters are basically syllable counting, with phrase-final stresses matched to strong beats in some genres (Dell & Halle 2009).

Since syllable weight plays an important role in English phonology and versification,[4] it may seem surprising that all attempts to create wholly or partly weight-based meters in English have failed. Not that these solo efforts produced bad poetry: On the contrary, Philip Sidney's quantitative hexameters and elegiac distichs in *Old Arcadia* are quite attractive, and Gerard Manley Hopkins's Sprung Rhythm, a meter of his own devising which is based on both stress and weight, is the vehicle of some of the most gorgeous verse the language has to offer.[5] The real reason that Sidney's and Hopkins's innovative weight-sensitive meters failed to gain traction is probably the phonological opacity of the English syllable weight distinction, due to its dual role in conditioning stress in the lexical phonology (Chomsky & Halle 1968) and being, in turn, modified by stress-conditioned resyllabification in the postlexical phonology (Kahn 1976). Generative phonology reveals the phonological rationale behind the quantitative experiments, and vindicates them against complacent critics who denied the phonological status of syllable quantity in English[6] and blamed the poets for being confused.[7]

Most quantitative meters select one or more constraints from the following menu:[8]

(5a) Strong positions
1. Must be a bimoraic foot ⏝ (less restrictive)
2. Must be a bimoraic syllable – (more restrictive)

(5b) Weak positions
1. Cannot be a bimoraic foot ⏝ (more restrictive)
2. Cannot be a bimoraic syllable – (less restrictive)

Note that a meter in which both examples 5a2 and 5b1 are enforced is isosyllabic.

[4] See below on W-resolution, which makes the moraic trochee the maximal occupant of a metrical position, a quantitative principle that remained alive from *Beowulf* through the twentieth century.

[5] The rest of this paragraph draws on Kiparsky (1989) for Sprung Rhythm and Hanson (2001) for Sidney's quantitative verse.

[6] "Since the would-be quantitative poet was obliged to remember constantly the arbitrarily assigned 'quantities' of the English syllables he chose to use, quantitative composition was a laborious academic-theoretical business, like all such nonempirical enterprises more gratifying to the self-congratulating practitioner than to the perplexed reader" (Fussell 1979, p. 68). "The rules that the poets used for determining syllable quantity applied to the written text, but not necessarily to what one would hear. . . . We must conclude that the quantitative experiment is somewhat like a written code—one needs to count and measure letters in order to determine the system" (Hollander 1985, pp. 65–66). "Length is Modern English is phonetic rather than phonemic" (Steele 1999, p. 267).

[7] "This atmosphere of confusion and muddlement enwraps almost all those who commit whoredom with this enchantress [quantitative meter]," "a curious measles or distemper which, dangerously but not by any means without beneficial results, affected English poetry. . .for more than half a century" (Saintsbury 1908, p. 168), "unnatural, arbitrary, inconsistent, and dependent on spelling" (Attridge 1974). To his credit, Attridge recognized that he tried to base his quantitative verse on the actual pronunciation of English, understood the distinction between weight and stress, was aware of the role of quantity in English, and tried to align stress with the ictus.

[8] In examples 5 and 6, ◡ stands for a light syllable, – for a heavy syllable, and ⏝ for a heavy syllable or two light syllables.

The various binary meters of Greek and Latin exploit all these options (West 1982, pp. 88–93; Kiparsky 2018b):

(6a) Strict iambic verse, with feet of the form ⏑ – (examples 5a2, 5b1)
(6b) Iambic with resolution in Strong, with feet of the form ⏑ ⏖ (examples 5a1, 5b1)
(6c) Iambic with resolution in Strong and split Weak, with feet of the form ⏕ ⏖ (examples 5a1, 5b2)

In ternary quantitative meters, both Strong and Weak are bimoraic (moraic trochees), and the correspondence constraints on positions determine the distribution of their monosyllabic and disyllabic realizations.

Tonal meters are widespread in Southeast Asia. A typical form is the popular Vietnamese Luc Bat "six–eight" stanza, with alternating six-syllable and eight-syllable lines. Words fall into two tonal classes, flat (*bằng*) and sharp (*trắc*). Flat words have either no tone or a low falling tone. Sharp words can have one of the tones that have a high component: *sắc*, *hỏi*, *ngã*, or *nặng*. Example 7 is from *Kim Vân Kiều* 'The Tale of by Kiều' by Nguyễn Du (translated by Lê Xuân Thuy):

(7) Trăm năm, trong cõi người ta,
Chữ tài, chữ mệnh, khéo là ghét nhau.
Trải qua một cuộc bể dâu,
Những điều trông thấy mà đau đớn lòng;
Lạ gì bỉ sắc, tư phong,
Trời xanh quen thói má hồng đánh ghen.

Within the span of hundred years of human existence,
what a bitter struggle is waged between genius and destiny!
How many harrowing events have occurred while mulberries cover the conquered sea!
Rich in beauty, unlucky in life!
Strange indeed, but little wonder,
since casting hatred upon rosy cheeks is a habit of the Blue Sky.

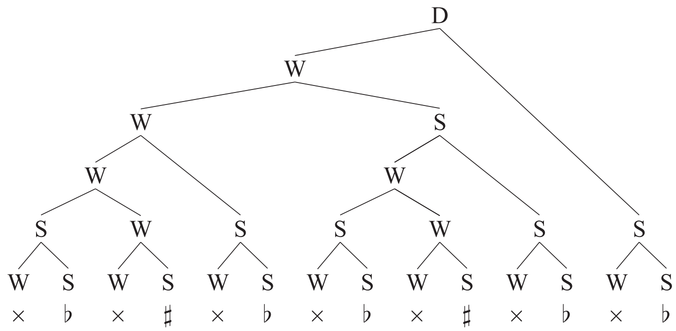
The end of an eight-syllable line rhymes with the end of the next six-syllable line and with the sixth syllable of the next eight-syllable line (rhymes underlined in example 7). Two words rhyme if both have sharp or flat tone, and identical or similar codas (rhymes with identical nuclei and codas are considered perfect). Even-numbered syllables, which I assume to be the heads of binary feet, are tonally restricted as shown. The schema is shown in example 8, where $\flat$ stands for flat, $\sharp$ for sharp, and $\times$ for unspecified:[9]

(8)
$\times \quad \flat \quad \times \quad \sharp \quad \times \quad \flat_a$
$\times \quad \flat \quad \times \quad \sharp \quad \times \quad \flat_a \quad \sharp \quad \flat_b$
$\times \quad \flat \quad \times \quad \sharp \quad \times \quad \flat_b$
$\times \quad \flat \quad \times \quad \sharp \quad \times \quad \flat_b \quad \sharp \quad \flat_c$

I posit the following structure for a distich (D), with the rule that the head of a Strong foot is $\flat$ and the head of a Weak foot is $\sharp$, and that the Strongest positions (6 and 8) rhyme:

[9]The schema is adopted from **http://en.wikipedia.org/wiki/Lục_bat**, where more references are given. An exception mentioned there is that the second syllable in odd lines is free when there is a break after the third.

(9)



3. ENGLISH METER

The meter of English is based on stress (some varieties of it also on syllable weight, e.g., Hopkins's Sprung Rhythm; Kiparsky 1989). Stress affects the temporal organization of the utterance and provides the inflection points for intonation contours. The local cues for stress in English are pitch, duration, loudness, and the distribution of certain segmental phonological features, such as vowel reduction in unstressed syllables and tapping of coronal stops before them (e.g., *atom* [ǽɾəm], *atomic* [ətʰámɨk]).

Like quantitative meters, stress-based meters can require either prominence in Strong positions or nonprominence in Weak positions, or both. Prominence in stress-based meters is usually defined by stress peaks, syllables bearing the strongest stress in some domain—a foot, a word, or a phrase. For instance, *philósophy* and *philósophìze* have one peak (underlined) in each domain, *philósophìzing* has two foot peaks but still just one word peak, and *tìme pást* has only a phrasal peak:

(10a) Constraint on Strong Positions: A Strong position must contain a stress peak.
(10b) Constraint on Weak Positions: A Weak position cannot contain a stress peak. Verse-initial and phrase-initial positions may be exempt ("inversion").

Popular meters of English obey example 10a. The standard literary binary meters, notably iambic pentameter, do not obey it. Instead, they impose example 10b, varying in what counts as a peak. For example, Milton's verse restricts phrasal peaks, while Shakespeare's restricts word peaks (Kiparsky 1977).

Microvariation in the correspondence constraints and in the prosodic phonology of poetic language generates a vast repertoire of metrical "dialects" or "styles," especially in major meters like iambic pentameter. One of the richest is the blank verse of Shakespeare's plays.[10]

The basic iambic pentameter template is example 11:

(11) Shakespeare's iambic pentameter
(11a) The basic pattern includes five iambic (Weak–Strong) feet, hierarchically organized as in example 19a.
(11b) At the end of a verse and before a phrase boundary, there can be an additional Weak position (an extrametrical syllable), shown by a slur.

[10]On the stricter, simpler form of iambic pentameter of his sonnets and narrative poems, see Hanson (2006).

The principal correspondence constraints are those in example 12:

(12a) Constraint on Positions: A position contains one syllable (obeyed in standard binary meters of English, except under the conditions specified below).
(12b) Constraint on Weak Positions: A Weak position cannot contain a stress peak (obeyed in standard binary meters of English, except for verse-initial and phrase-initial positions).

Compare the passage from Shakespeare in example 13a with its constructed unmetrical counterpart in example 13b, obtained by replacing monosyllabic stresses by polysyllabic ones (underlined) to create violations of the Constraint on Weak Positions:

(13a) And thou, / thrice-crow/nèd Queen/ of Night, / survey
With thy / chaste eye, / from thy / pale sphere / above,
Thy hun/tress' name / that my / full life / doth sway.
As You Like It, 3.2.2–4

(13b) *And thou, / well-main/<u>tain'd</u> Queen / of Night, / survey
*With con/<u>cealed</u> eyes, / from re/<u>mote</u> /spheres above,
*Thy hun/tress' name / that en/<u>tire</u> lives / doth sway.
(construct)

In the blank verse of Shakespeare's plays, optional elision and resolution allow a sequence of syllables to occupy a single position. Elision is the phonological reduction of two syllables to one, either by syncope or by contraction:

(14) Syncope elides a short vowel before a sonorant between a stressed syllable and an unstressed syllable within the same word.

(14a) I speak / not as / in ab/<u>solute</u> fear / of you.
(*Macbeth*, 4.3.39)

(14b) And see/ing ig/<u>norance</u> is / the curse / of God
(*Henry VI*, *Part II*, 4.7.72)

(14c) Upon / thy eye-/balls mur/<u>derous</u> ty/ranny
(*Henry VI*, *Part II*, 3.2.55)

(14d) That good / Duke Humph/rey trai/<u>torous</u>ly / is murder'd
(*Henry VI*, *Part II*, 3.2.124)

Syncope usually does not apply at the end of a line or before a break. It remains a live process in modern English, as in *infinite* → *inf'nite* or *gén(e)rative* versus *géneràte*:

(15) Contraction elides an unstressed vowel after a stressed one.[11]

(15a) Then, <u>seeing</u> / 'twas he / that made / you to / depose
(*Henry VI*, *Part III*, 1.2.26)

[11] A few words allow contraction across *v*: *ne'er*, *o'er*, *e'en*, but not **cle'er*, **clo'er*, **bea'er*.

(15b) Thou goest / to Co/ventry, / there to / behold
(*Richard II*, 1.2.258)

(15c) A sooth/sayer bids you / beware / the Ides / of March
(*Julius Caesar*, 1.2.19)

While these two elision rules are optional processes of the poetic language itself, resolution is a metrical license located among the correspondence constraints, which allows the placement of two syllables in one position. Being a metrical license, (*a*) it is not reflected in actual pronunciation, (*b*) it is sensitive to metrical structure, applying to Strong but not Weak positions, and (*c*) it is restricted to specific meters. Shakespeare's dramatic verse uses two types of resolution: W-resolution and F-resolution (unmetrical constructs added to show the limits of the operation):

(16) W-resolution: A light stressed syllable followed by another syllable within the same word can occupy a single Strong metrical position.

(16a) {Tyranni/cal | *Identi/cal} power: / if he / evade / us there
(*Coriolanus*, 3.3.2)

(16b) Stoop with / oppres/sion of / {their prodi/gal | *majesti/cal} weight
(*Richard II*, 3.4.34)

(16c) Of god/like {ami/ty | *chasti/ty}; which / appears / most strongly
(*The Merchant of Venice*, 3.4.3)

(17) F-resolution: Two function words (most often a preposition + *the*) can occupy a single Weak position.

(17a) His fel/lowship / in the cause / against / your city
(*Titus Andronicus*, 5.2.13)

(17b) Ill in / myself / to see, / and in thee / seeing ill
(*Richard II*, 2.1.94)

Elision, resolution, and extrametricality may be combined to yield feet of four syllables, or even five, as in examples 18a and 18b, where the last foot has syncope, W-resolution, and an extrametrical syllable:

(18a) And take / my milk / for gall, / you mur/dering mini sters
(*Macbeth*, 1.5.49)

(18b) With a / discove/ry of / the in/finite flatte ries
(*Timon of Athens*, 5.1.36)

(18c) What's He/cuba / to him, / or he / to Hecuba
(*Hamlet*, 2.2.585)

(18d) Some griefs / are medi/cina/ble; that / is one of them
(*Cymbeline*, 3.2.33)

(18e) To call / for recompense. / Appear / it to / your mind
(*Troilus and Cressida*, 3.3.3)

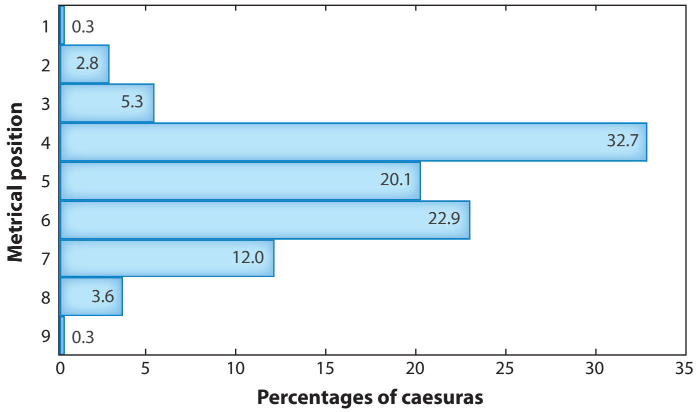


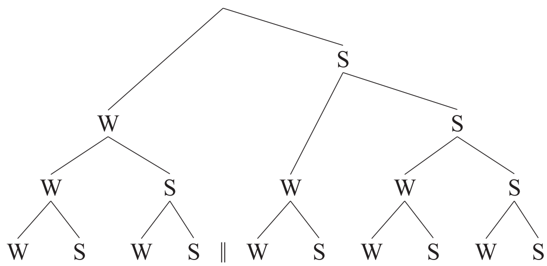
Figure 2

Percentages of caesuras after each metrical position in early iambic pentameter, $N = 358$. Based on samples from Shakespeare, Milton, Pope, and Wordsworth. Data from Keppel-Jones (2001, p. 234).

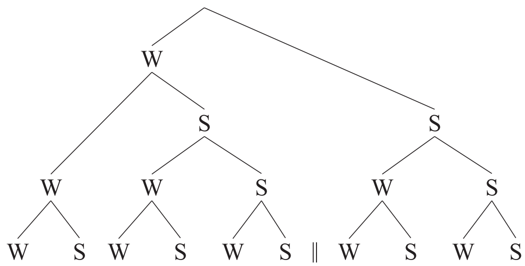
In Shakespeare's iambic pentameter, the feet are most commonly grouped as 2+3, with a caesura after the fourth syllable. **Figure 2** shows the percentages of caesuras after each position in early iambic pentameter.

In the later plays, the caesura after the sixth syllable becomes more common (Tarlinskaja 2014, p. 164). This suggests that the basic structure of Shakespeare's line shifts from example 19a to 19b:[12]

(19a)



(19b)



[12]The right branching in example 19a might reflect the general stylistic long-last preference, as in *friends, Romans, countrymen*; *let her rot, and perish, and be damned to-night*; *lands and revenues*; *drawn and ready*; *soft and delicate* (Ryan 2019).

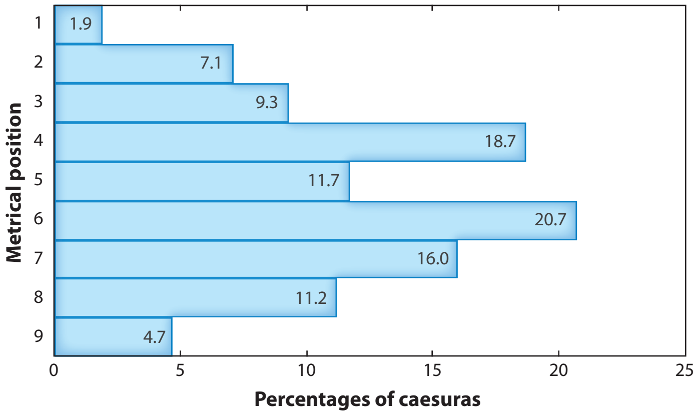


Figure 3

Percentages of caesuras after each metrical position in Browning's blank verse, $N = 321$. Based on "My Last Duchess" and "Return of the Druses" (beginning).

Varying the placement of the caesura in successive lines is an important feature of verse. In neoclassical iambic pentameter distichs (heroic couplets), a caesura after the fifth position becomes more common. Dividing the line into two constituents of equal size privileges the PARALLELISM CONSTRAINT, ubiquitous in meter and music (Lerdahl & Jackendoff 1983, p. 51), over STRICT LAYERING and ALIGNMENT. In the nineteenth century, the caesura shifts still further rightward; Browning favors it after the sixth beat in his early work (**Figure 3**) and after the seventh beat in his later work.

The descriptive generalizations outlined here beg for explanations. Why does W-resolution occur only in Strong positions and F-resolution only in Weak positions? The answer is that W-resolution in Weak positions would violate the CONSTRAINT ON WEAK POSITIONS, which is otherwise strictly obeyed. Example 20 presents a near-minimal pair that shows the contrast between elision and resolution:

(20) {A sooth/sayer | *A truth/teller} bids you / beware / the Ides / of March
(*Julius Caesar*, 1.2.19)

Conversely, F-resolution in Strong positions, as in construct 21b, would introduce a joint violation of both examples 10a and 12a:

(21a) Whom lep/rosy / o'ertake! — / i' the midst / o' the fight,
(*Antony and Cleopatra*, 3.10.11)

(21b) *Whom lep/rosy / take i' the / midst o' the / campaign
(construct)

Why are lexical stresses of polysyllabic words more restricted than stressed monosyllables? In particular, what makes them less suitable to appear in Weak positions? Lexical stresses of polysyllabic words are more salient than those of monosyllabic words, since they contrast with lexically unstressed syllables within the same word. By the same token, unstressed syllables of polysyllabic words are more saliently unstressed than those of monosyllabic words, and hence are even less suited to Strong positions, for they contrast with lexically stressed syllables within the same word. These considerations predict a hierarchy of prominence:

(22) $\acute{\mathrm{P}}$ a stressed syllable in a polysyllabic word (most prominent)
|
$\acute{\mathrm{M}}$ a stressed syllable in a monosyllabic word
|
$\overset{\times}{\mathrm{M}}$ an unstressed syllable in a monosyllabic word (necessarily a function word)
|
$\overset{\times}{\mathrm{P}}$ an unstressed syllable in a polysyllabic word (least prominent)

The following implicational relations are therefore predicted for constraints on metrical Strong and Weak positions (> means "implies"):

(23a) Constraints requiring stress in Strong positions: $\overset{\times}{\mathrm{P}} > \overset{\times}{\mathrm{M}} > \acute{\mathrm{M}} > \acute{\mathrm{P}}$

(23b) Constraints requiring absence of stress in Weak positions: $\acute{\mathrm{P}} > \acute{\mathrm{M}} > \overset{\times}{\mathrm{M}} > \overset{\times}{\mathrm{P}}$

Variation in English meter provides abundant support for this hierarchy. One example must suffice here: the types of extrametrical syllable permitted in different varieties of English iambic pentameter. The most restrictive system, found in early Marlowe (Schlerman 1989, p. 200), allows only unstressed syllables of polysyllabic words to be extrametrical, as in example 24a. In his later work, Marlowe adopts the commoner, less restrictive practice of allowing any unstressed syllable to be extrametrical, also observed by Shakespeare (see example 24b; Schlerman 1989, p. 202). Shakespeare's plays (but not his sonnets) extend it to secondary stresses of compound words, as in example 24c (Kiparsky 1977). The most unbuttoned of the Jacobean dramatists, such as Fletcher, put stressed full words, occasionally even phrasal peaks, into extrametrical positions (example 24d):

(24a) And sit / with Tam/burlaine / in all / his majestie
(*Tamburlaine*, 1.2.209)

(24b) I come / to bu/ry Cae/sar, not / to praise him.
Julius Caesar, 3.2.73

(24c) Quite o/verca/nopied / with lus/cious woodbine
(*A Midsummer Night's Dream*, 2.1.251)

(24d) But I / would reach you, / and bring / you to / your trot too.
(*The Tamer Tamed*, 1.3.9)

4. TEXT-SETTING

How is a metrical text fitted to the rhythmic pattern of a song or chant? The expectation is that prosodic features play a role in text-setting to the extent that they are important in meter and phonology. As mentioned above, these prosodic features are not necessarily phonemic, but they must function in the language's lexical phonology. Before proceeding to explore this conjecture, let us consider the relation between silence and phrasing in meter and music.

Structure above the line is organized primarily by two antagonistic constraints, PARALLELISM, which requires dividing a unit into two constituents of equal size, and CLOSURE, which requires making the final unit salient in some way. The normal way of achieving metrical closure is by making the final beat silent, which in song is realized by elongating the last overt one (Blumenfeld 2016).[13] But silent beats have other functions as well, which are of particular interest for the theory of meter.

[13]This is consistent with the closure effects mentioned in footnote 12, for there is no reason to posit silent feet in pentameters; in any case, the highest layer of constituency in blank verse is the line, so putting the shorter hemistich last gains no cohesion over putting the longer one last.

Let us begin with a simple example of a Hausa verse narrative documented by Schuh (1995).[14] The performer, the popular artist Ɗan Maraya Jos, accompanying himself on a plucked string instrument, uses three voices, a narrator (N) and two enacted characters, a woman (W, in boldface) and a fraudulent herbalist (H, in italics). As the extract in example 25 shows, the lines are of variable length, and no regular meter is discernible at first glance (macrons show vowel length; the other diacritics show tone):

(25)

N	Sai kàr ji dai mā́tar̃ gidā	Then you hear the woman of the house
N	Tā canḕ masà	She says to him
W	**“Wannàn àbu mài sauƙī kùwa**	**“That’s an easy thing**
W	**Bàri zân jē can kò cikin gidā.**”	**Let me go there to the house**.”
N	Tā kōmā́ can gidā	She goes back home
N	Jā́wō kwallā, sàyar̃	Drag out dress clothes, sell them
N	Jā́wō bōkitì mā, sàyar̃	Drag out buckets too, sell them
N	Tà tarar̃ tā mīƙà kuɗîn	She finds she hands over the money
N	An kai wâ bōkā nan dà nan	It’s taken to the herbalist immediately
N	Bōkā ya tattàrē	The herbalist collects it
N	Bōkā ya handàmē	The herbalist satisfies his desire
H	“*Jḕ ki, Allā́h zâi manà māgànī*.”	“*Off with you, Allah will be our remedy*.”

Once we understand the composition’s short lines as having empty metrical positions, its meter is instantly recognizable as regular anapestic tetrameter, a standard meter of Hausa:[15]

(26)

⏖	–	⏖	–	⏖	–	⏖	⏑
Sai	kàr	ji	dai	mā́	tar̃	gi	dā
Tā ca	nḕ	masà					
“Wan	**nàn**	**àbu**	**mài**	**sau**	**ƙī**	**kù**	**wa**
Bàri	**zân**	**jē**	**can**	**kò ci**	**kin**	**gi**	**dā**”
		Tā	kō	mā́	can	gi	dā
		Jā́	wō	kwal	lā,	sà	yar̃
	Jā́	wō	bō	kitì	mā,	sà	yar̃
		Tà ta	rar̃	tā	mī	ƙà ku	ɗîn
An	kai	wâ	bō	kā	nan	dà	nan
		Bō	kā	ya	tat	tà	rē
		Bō	kā	ya	han	dà	mē
Jē ki	*Al*	*lā́h*	*zâi*	*manà*	*mā*	*gà*	*nī*

The performance of the artist is the key to recognizing the meter. The empty positions in the meter correspond exactly to empty beats in the performance. After each sequence of missing beats, the singer picks up the regular metrical pattern. The empty beats are obvious from the accompaniment but cannot be parsed from the text alone, so the meter becomes intelligible only through the performance. The theoretical interest of this example is that it shows the metrical relevance of phonologically unrealized structure. It supports template-matching approaches over the parsing approach of Fabb & Halle (2008) and the “holistic” prosodic approach of Golston & Riad (2000) and Riad (2016). These theories operate on language directly and posit no independent pattern to supply the expectation of the missing beats.

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[14]See also **http://aflang.humanities.ucla.edu/language-materials/chadic-languages/hausa/hausa-poetry-song/** and Hayes & Schuh (2018).
[15]Here I diverge from Schuh’s (1995) analysis.

Empty beats challenge bottom-up parsing theories such as that of Fabb & Halle (2008), since one cannot parse what is not there. Their directional parsing algorithm also runs into difficulties in that the lines of this text would have to be scanned both left to right (for lines ending with empty beats) and right to left (for lines beginning with empty beats).[16] Moreover, the operations on grids that are their major means of negotiating mismatches cannot generate the short lines because the sequences of empty beats need not be constituents. Lines can begin or end with an odd number of empty beats, including sequences of three and five beats, which cannot be characterized as groups at any level, whether feet, dipods, or half-lines. But they are easily recognized in performance.

Empty beats also provide compelling arguments for including performance in the province of metrical theory, which Fabb & Halle (2008) explicitly reject. It turns out that line-medial empty beats occur only when the singer changes character—a salient event in performance but not a formal property of the text.

The prosodic metrical theory of Golston & Riad (2000) would probably have to deal with these data by different constraint rankings that output lines of various lengths, in this short sample alone of three, five, six, and seven metrical positions, both left-aligned and right-aligned. This would miss the generalization that lines of any length are admissible provided that they consist of a subsequence of the eight-syllable anapestic patterns.

Empty beats occur in Western art poetry as well.[17] Matthew Arnold's "Dover Beach" dramatizes a speaker expressing his thoughts to a companion as they view the Channel at night. Amidst the poem's subtly rhymed regular iambic pentameters are occasional shorter lines of four and three feet, placed to mark significant silences in the represented discourse. Example 27 reproduces lines 1–14, with ellipses to show where Samuel Barber's composition (Opus 3) puts a corresponding break in the short lines. Two such short lines begin the poem, conveying a sense of thoughts emerging from a contemplative silence. At line 9, there is again a missing beat, as the participants listen to the surf:

(27) … The sea is calm tonight. …
… The tide is full, the moon lies fair
Upon the straits; on the French coast, the light
Gleams and is gone; the cliffs of England stand,
Glimmering and vast, out in the tranquil bay.
Come to the window, sweet is the night-air!
Only, from the long line of spray
Where the sea meets the moon-blanched land,
Listen! … you hear the grating roar
Of pebbles which the waves draw back, and fling,
At their return, up the high strand,
Begin, and cease, and then again begin,
With tremulous cadence slow, and bring
The eternal note of sadness in. …

A purely formal analysis of the text as having a meter with variable line length misses the dramatic function of the short lines.

In setting English poems to music, most composers privilege the natural phonological phrasing, even at the expense of the poem's lineation and caesuras where they conflict. Phonological prominence also tends to trump metrical prominence, in that stressed syllables, especially of

[16] Note that it has what Fabb & Halle (2008) classify as a strict meter. Loose meters by definition are those which have at least one noniterative "parenthesis insertion" rule, which this meter does not have.
[17] There is a famous one in Hopkins's "Spelt from Sibyl's Leaves."

Figure 4

Sure On This Shining Night (from FOUR SONGS, OP. 13). Music by Samuel Barber, words by James Agee.

polysyllabic words, are preferentially aligned with Strong musical beats, regardless of whether they fall in Strong or Weak positions in the verse. Weak positions in iambic verse can therefore correspond to Strong musical positions, and conversely, especially if the phrasing and natural stress pattern require it (Halle & Lerdahl 1993, Hayes 2009).

A simple example is the first quatrain of James Agee's iambic "Sure on This Shining Night," which features a rhythmic parallelism of the couplets, with initial inversion in the odd lines:

(28) Sure on / this shi/ning night
Of star/-made sha/dows round,
Kindness / must watch / for me
This side / the ground.

Samuel Barber's composition *Four Songs* (**Figure 4**) highlights this parallelism, without attempting a close rendering of the metrical foot structure.

Dell & Halle (2009) show that while English matches stresses to strong musical beats across the board, French does so only at the ends of lines, and conversely that French traditional songs require a parallel pairing of syllables to beats in each stanza, which is not the case in English. They propose to derive both differences from the fact that stress in English is perceptually salient throughout the utterance, and only before major breaks in French, and that French meter is basically syllable counting.

For the rendering of stress and quantity it is difficult to give any general rules. It varies by style and genre even within a language and period. Finnish provides an illustration of this variability. In this language, the primary word stress falls on the initial syllable. Vowel and consonant length is contrastive throughout the word, as in *tule* 'come!', *tulee* 'comes', *tuule* 'blow!' (wind), *tuulee* 'blows', *tullee* 'probably comes', and *tuullee* 'it is probably windy'. But in singing, vowel length can be neutralized by lengthening, and stress can be displaced onto syllables that fall in prominent positions. In the following rendition of a traditional Finnish ballad (the counterpart of "Edward, Edward"), the singers lengthen the stressed open syllables, pronouncing, for instance, *túlet* 'you come' as *túulet*:[18]

(29) *Velisurmaaja* 'The Fratricide', sung by Niekku. 4-3 ballad meter.

1. Mis.täs tú.let kús.tas tú.let, — Where are you coming from,
2. Poi.ka.ni í.loi.nen? — My cheerful son?
3. Mí.e tú.len mé.ren rán.nast, — I'm coming from the seashore,
4. Miu ä́i.ti.ni kul.tai.nén. — My dear mother.

[18] **https://www.youtube.com/watch?v=NwAcHcuYzL4**.

When an unstressed syllable has to fall on a strong beat, they sometimes shift the word stress onto it, as in line 3 of this verse:

(30)	1.	Míst on míek.kais vér.re.hen túl.lu,	How did your sword get bloody,
	2.	Poi.ka.ni í.loi.nen?	My cheerful son?
	3.	Míe ta.póin van.hém.man vél.jen,	I killed my older brother,
	4.	Miu ǟ́i.ti.ni kul.tai.nén.	My dear mother.

Native Finnish meters are ingeniously adapted to the coexistence of phonemic vowel length and initial stress so as to jointly optimize variety and phonological faithfulness. The *Kalevala* meter consists of eight-syllable lines made up of four trochaic (Strong–Weak) feet, with obligatory alliteration and parallelism. The basic metrical rule is that a stressed (i.e., word-initial) syllable must be ⏑ in Weak positions and – in Strong positions (⏑ = CV̆). This avoids lengthening of vowels in the sensitive first syllable, which would neutralize the most important site of the length contrast. The resulting mismatches between stress and metrical positions in words with an odd number of syllables create the main rhythmic excitement of this meter.

In the rendition by Niekku, the singers foreground the tension between phonology and meter by resolving it differently in repetitions of lines 1 and 3. They sing them once with the original word stress, and the second time they shift the word stress to the syllable that falls on the metrical stress, in each case with lengthening of some of the stressed vowels:[19]

(31)	a1.	Ei pí.tǟi.si núo.ren néi.en	A young girl shouldn't
	a2.	Ei pi.tǟ́i.si núo.ren néi.en	A young girl shouldn't
	b.	Vás.ta.kás.va.van ká.na.sen	A chick still growing
	c1.	Ús.ko.a ú.ron sá.no.ja	Trust the words of a male
	c2.	Us.ko.á ú.ron sa.nó.ja	Trust the words of a male
	d.	Míe.hen váls.kin ván.non.noi.ta	The promises of a cheating man

In general, distinctive quantity seems to be reflected more faithfully in text-setting than stress is (Hayes 2016). Since syllable weight is the only active prosodic feature in the lexical phonology of Urdu (or, at least, by far the most important one), and its meters are also based on weight, good singers take care to preserve it. Begum Akhtar's ghazal renderings respect syllable weight as far as possible while maintaining Urdu's phonemic vowel length contrast, even at the cost of some compromises in the syllable count. Metrically prominent positions are sung as bimoraic. In words like *gul* 'rose', this is done by extending the note over the vowel and the sonorant coda. Words like *səb* 'all' are bimoraic and occupy heavy positions in the meter, but their closing obstruent cannot carry a note, nor can the phonemically short vowel be lengthened lest the length contrast is lost. In positions that require a heavy syllable, such words are sung with final *-ə* inserted as a last resort, which allows their heavy syllables that have only one singable mora to fill the musical space allotted to them. In this system, faithfulness to syllable weight trumps faithfulness to the syllable count.[20]

The investigation of text-to-tune alignment in crosslinguistic perspective is only beginning, but preliminary results already indicate that the negotiation of phonology and meter in song provides a sensitive probe into the prosodic organization of language.

[19] *Suomen kansanmusiikki* 1 (Folk Music of Finland 1), Kansanmusiikki-instituutti 1988.

[20] A fine example is Begum Akhtar's rendering of Ghalib's *Sab Kahan Kuch Lala-o-Gul Mein* (**https://www.youtube.com/watch?v=0wggu6H8Qxo**).

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